Quest 2: The Axeperiment

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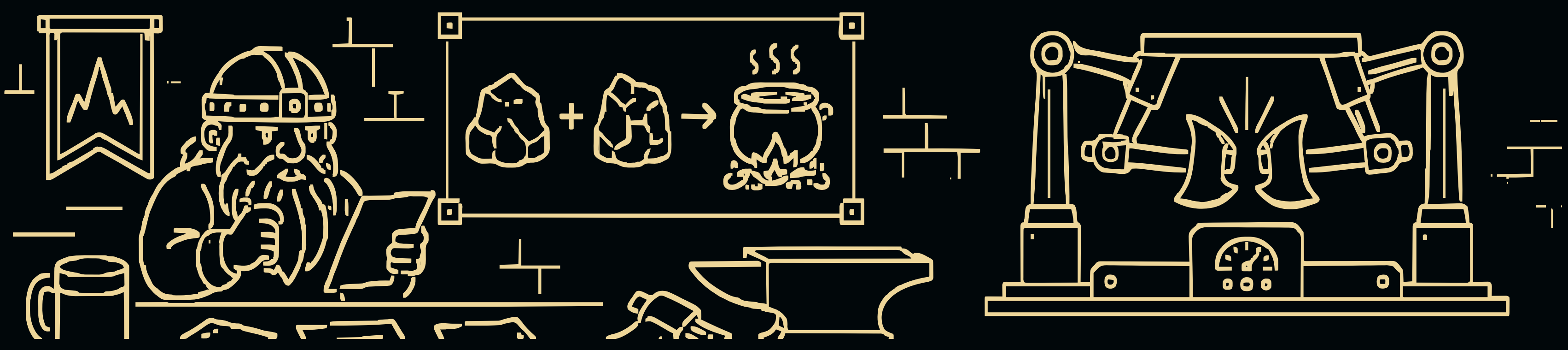

Part I

Story section

Once Quackshire had finally been secured, we wasted no time and set off for the Forge Mountain. Our next stop was the workshop of our old friend, the dwarven blacksmith Gimlock Ironstack.

Gimlock immediately asked where we were heading. I simply replied, "Nulldor." Then I pointed at the stone sticking out of Frodoff's backpack. He stared at it for a long moment, sighed deeply, and muttered, "Not this nonsense again." Without asking another question, he agreed to come.

There was just one small problem. Before leaving, Gimlock insisted on finishing his latest research into the hardness of several newly forged metal alloys. As the mountain's leading expert in the field, he explained that the procedure was extremely scientific: hit two axe blades together and see which one complains first. We offered to help him finish the analysis as quickly as possible.

Gimlock handed us his research notes (your notes). Each line described one group of alloys and contained **the results of comparing every possible pair of blades** forged from that group. For example, `3>5` meant that alloy `3` had proved harder than alloy `5`.

Using only these comparison results, we had to **determine the order** of the alloys in every group, from the hardest to the softest.

We then read the ordered identifiers as a single number for each group and added all of those numbers together to **produce the checksum.** Gimlock explained that the checksum would make the research much easier to verify if the original notes were ever misplaced and later reconstructed.

Example based on the following notes:

```
4<3 2<5 3>1 1>2 1<5 4<1 3>2 4<2 3>5 4<5
4>1 2>1 3>2 1<5 4>5 2<5 3>1 4>3 2<4 5<3
5>1 5>2 4>1 2<4 3>2 3<4 1>2 3>5 1<3 4>5
```

In the first group (the first line), the alloys could be arranged from the hardest to the softest as: `3 5 1 2 4`. This gave us the number `35124`.

In the second group, the correct order was: `4 3 5 2 1`. This gave us the number `43521`.

In the third group, the correct order was: `4 3 5 1 2`. This gave us the number `43512`.

The final checksum was therefore: 35124 + 43521 + 43512 = **`122157`**

What was the final checksum of the research notes?

Your notes for this part:

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Part I solved with answer: 330882

Check your progress

Part II

Gimlock's second notebook (your notes) looked almost identical to the first one, except that **each group was missing the result of exactly one comparison**. Apparently, forging a brand-new axe for every test had taken rather longer than expected. We suggested reusing an axe, but Gimlock called that "scientifically irresponsible."

He asked us to help him complete the missing measurements, but before either of us could answer, Frodoff's eyes suddenly turned golden. He studied the notes for a few moments, then smiled.

"Why bother making the last comparison?" he asked. "If alloy `1` was harder than alloy `2`, and alloy `2` was harder than alloy `3`, then alloy `1` had to be harder than alloy `3` as well."

Gimlock and I exchanged a glance. Following Frodoff's reasoning, we **deduced the missing comparison** in every group, arranged the alloys from the hardest to the softest, and **calculated the checksum exactly as before.** We had been rather lucky. A missing comparison cannot always be deduced from the others, but Frodoff's method was enough to solve every group in these notes.

Example based on the following notes:

```
2<5 3>1 1>2 1<5 4<1 3>2 4<2 3>5 4<5
2>1 3>2 1<5 4>5 2<5 3>1 4>3 2<4 5<3
5>1 5>2 4>1 3>2 3<4 1>2 3>5 1<3 4>5
```

This is the same example as in Part I, with exactly one comparison removed from each group. The missing comparison can still be deduced, so the alloy order, the resulting numbers, and the final checksum are identical.

The final checksum is therefore again **`122157`**.

What was the final checksum of the second set of research notes?

Your notes for this part:

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Part II solved with answer: 56830573962

Check your progress

Part III

Gimlock disappeared into his workshop and eventually emerged with one last bundle of notes (your notes), rescued from beneath several years' worth of perfectly organised dwarven paperwork. These were from a very early stage of his research, and this time every group was missing many comparison results.

With so **many measurements missing**, the deduction was no longer as obvious as before. Frodoff showed us that the remaining results, combined with the same reasoning, were enough to reconstruct the complete ordering of every alloy group from the hardest to the softest, and **calculated the checksum exactly as before.**

Example based on the following notes:

```
3>1 1>2 1<5 4<2 3>5
2>1 2<5 3>1 4>3 2<4 5<3
5>1 3<4 1>2 3>5
```

This is the same example as in the previous parts, with some comparisons removed from each group. The remaining results are enough to determine the alloy order, so the resulting numbers and the final checksum are identical.

The final checksum is therefore again **`122157`**.

What was the final checksum of the last set of research notes?

Your notes for this part:

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Part III solved with answer: 592639023780

Check your progress

Puzzle solved! Don't stop now!

Post your solution, compare ideas, and help others grow on Reddit or join us for a chat on Discord

Bonus for Supporters!

This feature is available only to Supporters as a small "thank you" for supporting this event. Click the link below to **get all 100 versions of the input notes prepared for this Quest**, along with the answers! Please don't host it anywhere; especially please don't add it to your solutions repo. Thank you for being awesome! ❤

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